

## Lab 5: Span, Linear Independence, and Subspaces

EECS 245, Fall 2026 at the University of Michigan

due by the end of your lab section on Wednesday, September 30th, 2026

Name: \_\_\_\_\_

username: \_\_\_\_\_

Each lab worksheet will contain several activities, some of which will involve writing code and others that will involve writing math on paper. To receive credit for a lab, you must complete Activities 1-4 and show your lab TA by the end of the lab section.

While you must get checked off by your lab TA **individually**, we encourage you to form groups with 1-2 other students to complete the activities together.

### Recap: Span, Linear Independence, Subspaces, Bases, and Dimension

- ([Chapter 4.1](#)) The **span** of a set of vectors is the set of all possible linear combinations of the vectors in the set.

$$\text{span}(\{\vec{v}_1, \vec{v}_2, \dots, \vec{v}_d\}) = \{a_1\vec{v}_1 + a_2\vec{v}_2 + \dots + a_d\vec{v}_d \mid a_1, a_2, \dots, a_d \in \mathbb{R}\}$$

- The span of one vector in  $\mathbb{R}^n$  is a line through the origin.
  - The span of two **non-parallel** vectors in  $\mathbb{R}^n$  is a plane through the origin; this plane is called a **2-dimensional subspace** of  $\mathbb{R}^n$ .
  - In general, the span of  $d$  vectors in  $\mathbb{R}^n$  is a subspace of  $\mathbb{R}^n$  of dimension 0 to  $d$ , depending on the vectors and their relationships.
  - Think of a  $d$ -dimensional subspace of  $\mathbb{R}^n$  as a “slice” of  $\mathbb{R}^n$  that goes through the origin, in which you can move in  $d$  directions.
- ([Chapter 4.2](#)) Intuitively, a set is **linearly independent** if none of the vectors in the set is a redundant building block: you can’t create it from the other vectors.

Formally,  $\vec{v}_1, \vec{v}_2, \dots, \vec{v}_d$  are **linearly independent** if the only solution to

$$a_1\vec{v}_1 + a_2\vec{v}_2 + \dots + a_d\vec{v}_d = \vec{0}$$

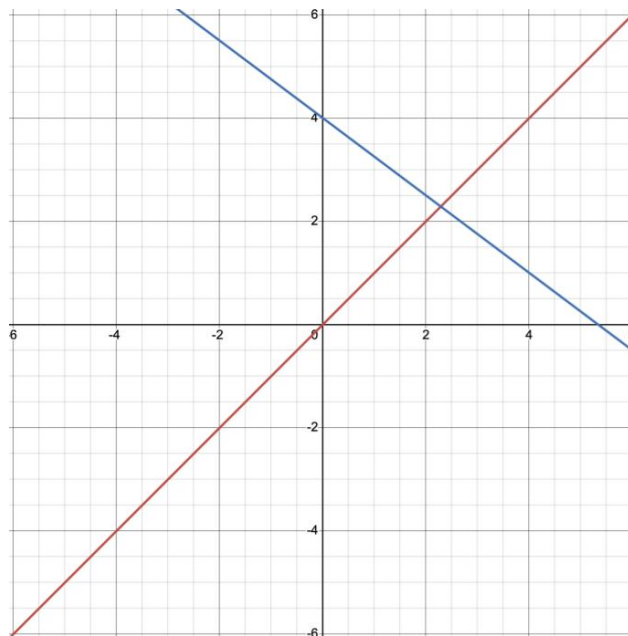
is  $a_1 = a_2 = \dots = a_d = 0$ . If a set of vectors is not linearly independent, it is **linearly dependent**.

- (Chapter 4.3) A **subspace**  $S$  of a vector space  $V$  is a set of vectors where:

1.  $\vec{0} \in S$
2.  $\vec{u}, \vec{v} \in S \rightarrow \vec{u} + \vec{v} \in S$
3.  $\vec{u} \in S, c \in \mathbb{R} \rightarrow c\vec{u} \in S$

If you take any two vectors  $\vec{u}, \vec{v} \in S$ , then any linear combination  $c\vec{u} + d\vec{v}$  must also be in  $S$ .

- As an example, let's consider  $\mathbb{R}^2$ , which itself is a vector space.



- The line through the origin is a subspace of  $\mathbb{R}^2$ , with dimension 1. It is the span of the vector  $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ .
- The other line, however, is **not** a subspace of  $\mathbb{R}^2$ , since it doesn't pass through the origin.
- A **basis** for a subspace  $S$  is a set of vectors that:
  1. span all of  $S$
  2. are linearly independent

A basis for a subspace is a minimal set of vectors that spans the whole subspace. All subspaces have infinitely many bases. For example,  $\left\{\begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix}\right\}$  and  $\left\{\begin{bmatrix} 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ 3 \end{bmatrix}\right\}$  are both bases for  $\mathbb{R}^2$ .

- The **dimension** of a subspace  $S$ , denoted  $\dim(S)$ , is the number of vectors in any basis for  $S$ .

### Activity 1: Formal Definition of Linear Independence

Suppose  $\vec{v}_1, \vec{v}_2, \dots, \vec{v}_d \in \mathbb{R}^n$ , and that  $\vec{b} \in \text{span}(\{\vec{v}_1, \vec{v}_2, \dots, \vec{v}_d\})$ .

- a) Give a one sentence English explanation of what it means for  $\vec{b} \in \text{span}(\{\vec{v}_1, \vec{v}_2, \dots, \vec{v}_d\})$ .

- b) Suppose that  $a_1\vec{v}_1 + a_2\vec{v}_2 + \dots + a_d\vec{v}_d = \vec{b}$  **and**  $c_1\vec{v}_1 + c_2\vec{v}_2 + \dots + c_d\vec{v}_d = \vec{b}$ , where at least one of the  $a_i$ 's is different from its corresponding  $c_i$ .

Using the formal definition of linear independence from [Chapter 4.2](#), determine whether or not  $\vec{v}_1, \vec{v}_2, \dots, \vec{v}_d$  are linearly independent, and prove your answer.

- c) Find another set of coefficients  $k_1, k_2, \dots, k_d$  such that

$$k_1\vec{v}_1 + k_2\vec{v}_2 + \dots + k_d\vec{v}_d = \vec{b}$$

and at least one of the  $k_i$ 's is different from its corresponding  $a_i$  or  $c_i$ .

By doing this, you're showing that if there is at least one way to write  $\vec{b}$  as a linear combination of a set of vectors, then there are infinitely many ways to write  $\vec{b}$  as a linear combination of those vectors; there can't just be two or three ways to do it.

## Activity 2: Thinking in Higher Dimensions

a) Suppose  $\vec{v}_1, \vec{v}_2, \dots, \vec{v}_8$  are 8 vectors in  $\mathbb{R}^5$ . Fill in each blank below with one of the provided options, and explain your reasoning.

(i) These vectors \_\_\_\_\_ span all of  $\mathbb{R}^5$ .  
(options: do, do not, might)

(ii) These vectors \_\_\_\_\_ linearly independent.  
(options: are, are not, might be)

(iii) Any 5 of these vectors \_\_\_\_\_ span all of  $\mathbb{R}^5$ .  
(options: do, do not, might)

b) Suppose  $\vec{u}_1, \vec{u}_2, \dots, \vec{u}_{10}$  are 10 non-zero vectors in  $\mathbb{R}^{11}$ .

Furthermore, suppose that  $\text{span}(\{\vec{u}_1, \vec{u}_2, \dots, \vec{u}_{10}\})$  is a 6-dimensional subspace of  $\mathbb{R}^{11}$ . This means that there exists a subset of 6 of these vectors that is linearly independent and spans the same 6-dimensional subspace as the original 10 vectors; we just don't know which 6.

(i) Let  $k$  be the dimension of the subspace spanned by a subset of 4 of these vectors. What are all possible values of  $k$ ?

(ii) Let  $m$  be the dimension of the subspace spanned by a subset of 7 of these vectors. What are all possible values of  $m$ ?

### Activity 3: Finding a Linearly Independent Subset

Recall from [Chapter 4.2](#) that a set of vectors  $\vec{v}_1, \vec{v}_2, \dots, \vec{v}_d$  is **linearly independent** if either of the following equivalent conditions hold:

- None of the vectors can be written as a linear combination of the others.
- The only way to create the zero vector as a linear combination of the vectors is if all the coefficients are zero. In other words, the only solution to

$$a_1\vec{v}_1 + a_2\vec{v}_2 + \dots + a_d\vec{v}_d = \vec{0}$$

is  $a_1 = a_2 = \dots = a_d = 0$ .

[Chapter 4.2](#) introduces an algorithm for finding a linearly independent subset of a given set of vectors with the same span as the original set:

```
given v_1, v_2, ..., v_d
initialize linearly independent set S = {v_1}
for i = 2 to d:
    if v_i is not a linear combination of S:
        add v_i to S
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In each of the parts below, find a **linearly independent** set of vectors that spans the same span as the given set of vectors. There are multiple possible answers for each part.

a)

$$\vec{v}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \quad \vec{v}_2 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \quad \vec{v}_3 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \quad \vec{v}_4 = \begin{bmatrix} 2 \\ 3 \\ 4 \end{bmatrix}$$

**b)**

$$\vec{v}_1 = \begin{bmatrix} 1 \\ -1 \\ 0 \\ 0 \end{bmatrix}, \quad \vec{v}_2 = \begin{bmatrix} 1 \\ 0 \\ -1 \\ 0 \end{bmatrix}, \quad \vec{v}_3 = \begin{bmatrix} 1 \\ 0 \\ 0 \\ -1 \end{bmatrix}, \quad \vec{v}_4 = \begin{bmatrix} 0 \\ 1 \\ -1 \\ 0 \end{bmatrix}, \quad \vec{v}_5 = \begin{bmatrix} 0 \\ 1 \\ 0 \\ -1 \end{bmatrix}, \quad \vec{v}_6 = \begin{bmatrix} 0 \\ 0 \\ 1 \\ -1 \end{bmatrix}$$

*Hint: Use the 0's in the vectors strategically, plus use the fact that you can't have more than 4 linearly independent vectors in  $\mathbb{R}^4$ .*

#### Activity 4: Finding Bases for Subspaces

In each part below, find **two different possible bases** for the given subspace, and state the **dimension** of the subspace.

a)  $S = \text{span} \left( \left\{ \begin{bmatrix} 1 \\ 3 \\ 3 \end{bmatrix}, \begin{bmatrix} -3 \\ -9 \\ -9 \end{bmatrix}, \begin{bmatrix} 1 \\ 5 \\ -1 \end{bmatrix}, \begin{bmatrix} 2 \\ 7 \\ 4 \end{bmatrix}, \begin{bmatrix} 1 \\ 4 \\ 1 \end{bmatrix} \right\} \right)$

b)  $S = \left\{ \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} \mid v_1 = -v_2; v_1, v_2 \in \mathbb{R} \right\}$

c)  $S = \left\{ \begin{bmatrix} v_1 \\ v_2 \\ v_3 \\ v_4 \end{bmatrix} \mid v_4 = 0; v_1, v_2, v_3 \in \mathbb{R} \right\}$

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The rest of this worksheet is extra practice. Don't feel pressured to answer all of these problems in lab, but make sure to attempt them at some point.

### Activity 5: Introduction to Subspaces

Only one of the following is a subspace of  $\mathbb{R}^3$ . Which one? Explain why the others are not subspaces.

The set of vectors  $\vec{v} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$  in  $\mathbb{R}^3$  such that

(i)  $x + 2y - 3z = 4$

(ii)  $\vec{v}$  is on the line  $L = \begin{bmatrix} 1 \\ -2 \\ 0 \end{bmatrix} + t \begin{bmatrix} 2 \\ 3 \\ 4 \end{bmatrix}, t \in \mathbb{R}$

(iii)  $x + y + z = 0$  and  $x - y + z = 1$

(iv)  $x = -z$  and  $x = z$

(v)  $x^2 + y^2 = z$

### Activity 6: Finding Non-Examples of Subspaces

In this activity, you'll find sets of vectors in  $\mathbb{R}^2$  that satisfy some, but not all, of the requirements for a subspace. Think creatively, and since we're working in  $\mathbb{R}^2$ , visualize the vectors!

- a) Find a set of vectors in  $\mathbb{R}^2$  such that the sum of any two vectors  $\vec{u}$  and  $\vec{v}$  in the set is also in the set, but  $\frac{1}{2}\vec{v}$  is possibly not in the set.

- b) Find a set of vectors in  $\mathbb{R}^2$  such that  $c\vec{v}$  is in the set for any vector  $\vec{v}$  in the set and any scalar  $c$ , but the sum of any two vectors  $\vec{u}$  and  $\vec{v}$  in the set is possibly not in the set.